

Row V, second part: 2b completed, and a check of (1.11)–(1.16)

The UV extracted in $\overline{\text{MS}}$, the p^+ integral done, and one correction to what I told you before

$$\eta = \frac{p^+}{k^+}, \quad \bar{\eta} = 1 - \eta, \quad L = \log \frac{k^+}{\Lambda}, \quad \bar{\mu}^2 = 4\pi\mu^2 e^{-\gamma_E}, \quad C_{\text{UV}}(\eta) = \eta\bar{\eta} + \frac{\eta}{\bar{\eta}} + \frac{\bar{\eta}}{\eta} = \frac{\mathcal{P}_{gg}(\eta)}{2N_c}, \quad \mathcal{W} = \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2}$$

(1.11) through (1.16) are all correct — the six tensor blocks, the prefactor, the $\delta^{(2)}$, the w integration, the six p^+ weights, the colour rewriting, the add-and-subtract and the split. Only presentational fixes remain (§1). **Part 2b is completed in §3.** Its UV pole carries the two moments

$$\int_{\lambda}^{1-\lambda} C_{\text{UV}} d\eta = 2L - \frac{11}{6}, \quad \int_{\lambda}^{1-\lambda} C_{\text{UV}} \log \bar{\eta}^2 d\eta = -L^2 + \frac{67}{18} - \frac{\pi^2}{3},$$

so you get a $\frac{2L}{\epsilon}$, the β_0 -type constant $-\frac{11}{6\epsilon}$, and a $-\log^2 \frac{k^+}{\Lambda}$.

A correction to what I said last time. I called the large- $|z|$ logarithm of this row a genuine infrared divergence needing cancellation against $\bar{\mathfrak{B}}_2$. That was computed with the phase dropped. With the phase kept, the z -dependence is $e^{i\bar{\eta}k \cdot z}$, so the large- $|z|$ region is cut at $|z| \sim \frac{1}{\bar{\eta}k_{\perp}}$ and there is **no infrared divergence at all** — it becomes $\log \bar{\eta}^2 k_{\perp}^2$, and that is precisely what turns into the $-\log^2 \frac{k^+}{\Lambda}$ above. The only pole in this row is the ultraviolet one at $z \rightarrow x$.

1. Check of (1.11)–(1.16): the algebra is right

Everything below was verified numerically (details in the scripts):

- **(1.11)** — all six tensor blocks reproduce the $\mathfrak{C}_1 \otimes \mathfrak{B}_2$ contraction $\delta_{im} \text{tr} B - \frac{k^+}{p^+} B_{mi} - \frac{k^+}{k^+ - p^+} B_{im}$ exactly (2.8×10^{-14} , $d_{\perp} = 2, 3, 4, 5$); the prefactor $\frac{g^4}{8\pi^4} \frac{p^+(k^+ - p^+)}{k^+}$ is exact; the $\delta^{(2)}$ and $\frac{(z-x)^m}{(z-x)^2}$ follow from $\mathfrak{C}_1$'s arguments.
- **(1.12)** — the $\delta^{(2)}$ sets $w = z - \frac{p^+}{k^+}(z - x)$ and gives exactly the two phases; all six p^+ weights transfer correctly.
- **(1.13)** — $f^{ebf} = -f^{fbe}$ and $-f^{abc} f^{cbd} U^{df} = +N_c U^{af}$, both exact.
- **(1.14)** — trivially exact, and the identity that licenses it, $f^{abc} f^{def} U^{be}(z) U^{cf}(z) = N_c U^{ad}(z)$, is correct with $S^{ad} = U^{ad}$.
- **(1.15)/(1.16)** — $\mathcal{C}_{2a} + \mathcal{C}_{2b}$ reproduces (1.13)'s colour identically (2.2×10^{-15}).

Four things to fix, all presentational but all real:

1. **The phases sit outside the integrals they depend on** in (1.12)–(1.16). $e^{-ik \cdot (w' - z)} e^{-ik \cdot (z - x) p^+ / k^+}$ is written before the $\left\{ \right.$, i.e. outside both $\int_{\Lambda}^{k^+ - \Lambda} dp^+$ and $\int_{x, y, z, x'}$, yet it depends on z, x and p^+ . Move it inside. (Cleanest: write it once as $\exp[-ik \cdot (w' - \bar{\eta}z - \eta x)]$ — see §2.)
2. **The colour factors and ρ 's sit outside** $\int_{x, y, z, x'}$: the first $\left\{ \right.$ closes right after the tensor bracket. It belongs at the very end.
3. **Braces unbalanced** in (1.12)–(1.16): one $\left\{ \right.$ opened, two $\left. \right\}$ closed. Fixing (2) fixes this.
4. Write $2a$'s colour as $-f^{adc} f^{fbe} U^{db}(x) [U^{ce}(z) - U^{ce}(x)]$ (exact, same object). Then its UV finiteness is visible on the page instead of needing an argument.

2. Setting up the z integral of 2b

In 2b the colour factor $N_c [U^{af}(y) - U^{af}(x)]$ is **z -independent**, so the whole z -dependence is the kernel and the phase. Write the total phase in the form that makes this transparent:

$$e^{-ik \cdot (w' - z)} e^{-i\eta k \cdot (z - x)} = \exp[-ik \cdot (w' - \bar{\eta}z - \eta x)] = e^{-ik \cdot w'} e^{i\bar{\eta}k \cdot z} e^{i\eta k \cdot x}.$$

From §5 of the earlier note, the braces of (1.16) behave near $z \rightarrow x$ as

$$\xrightarrow{z \rightarrow x} \frac{C_{UV}(\eta)}{k^+} \mathcal{W} \frac{1}{(z-x)^2}, \quad \mathcal{W} = \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2},$$

with all the $d_\perp$'s cancelling out of the residue. That last point matters here: because the residue carries no $d_\perp$, there is no $O(\epsilon)$ piece of it to multiply the $\frac{1}{\epsilon}$, so no extra finite term sneaks in from the $d_\perp$ bookkeeping.

The master integral

The UV counterterm's z integral is, with $q = \bar{\eta} k$,

$$\int d^2 z \frac{e^{iq \cdot (z-x)}}{(z-x)^2} = 2\pi \int_\rho^\infty \frac{dr}{r} J_0(qr) = -\pi \log \left[\frac{q^2 \rho^2 e^{2\gamma_E}}{4} \right] \quad \Rightarrow \quad -\pi \left[\frac{1}{\epsilon} + \log \frac{q^2}{\bar{\mu}^2} \right]$$

in $d_\perp = 2 - 2\epsilon$ with $\overline{\text{MS}}$. Checked against the Bessel integral:

ρ	q	numerical	$-\pi \log \frac{q^2 \rho^2 e^{2\gamma_E}}{4}$
10^{-4}	1.30	56.95021122	56.9502112
10^{-6}	1.30	85.88534885	85.88534885
10^{-6}	0.40	93.29105661	93.29105661

The oscillation converges the integral at large r on its own, so this $\frac{1}{\epsilon}$ is purely ultraviolet — which is the correction flagged at the top.

3. The p^+ integral: two moments of C_{UV}

With $\int_\Lambda^{k^+ - \Lambda} \frac{dp^+}{2\pi} = \frac{k^+}{2\pi} \int_\lambda^{1-\lambda} d\eta$, $\lambda = \frac{\Lambda}{k^+}$, everything reduces to two integrals of C_{UV} :

$$I_0 = \int_\lambda^{1-\lambda} C_{UV}(\eta) d\eta = \underbrace{\frac{1}{6}}_{\eta\bar{\eta}} + \underbrace{(L-1)}_{\eta/\bar{\eta}} + \underbrace{(L-1)}_{\bar{\eta}/\eta} = \boxed{2L - \frac{11}{6}}$$

$$I_1 = \int_\lambda^{1-\lambda} C_{UV}(\eta) \log \bar{\eta}^2 d\eta = \underbrace{-\frac{5}{18}}_{\eta\bar{\eta}} + \underbrace{(-L^2 + 2)}_{\eta/\bar{\eta}} + \underbrace{\left(2 - \frac{\pi^2}{3}\right)}_{\bar{\eta}/\eta} = \boxed{-L^2 + \frac{67}{18} - \frac{\pi^2}{3}}$$

Λ/k^+	I_0 quadrature	I_0 formula	I_1 quadrature	I_1 formula
10^{-3}	11.98418	11.98218	-47.29853	-47.28473
10^{-5}	21.19254	21.19252	-132.11533	-132.11510
10^{-8}	35.00803	35.00803	-338.88913	-338.88912

The $-\frac{11}{6}$ is β_0 . $\frac{11N_c}{6}$ is exactly the $\delta(1-z)$ coefficient of $\mathcal{P}_{gg}$, i.e. $\frac{\beta_0}{2}$ at $n_f = 0$. Two turns ago I said that constant would have to come from the *virtual* rows, where the soft endpoint is cut by transverse rather than rapidity kinematics — and this is a virtual row, and here it is, sitting on the $\frac{1}{\epsilon}$. Its partner $-\frac{2n_f}{3}$ must come from the $g \rightarrow q\bar{q}$ row; once you have that one, check that the two combine into β_0 , and this pole renormalises the coupling rather than needing a fragmentation function. The $\log \frac{k_\perp^2}{\bar{\mu}^2}$ riding alongside is then the $\alpha_s(k_\perp)$ we were looking for.

4. The completed part 2b

$$\left. \frac{d^3 N_{2b}^{\text{NLO}}}{d^3 k} \right|_{\text{UV}} = -\frac{1}{(2\pi)^3} \frac{g^4}{16\pi^4} \left\{ \left[2L - \frac{11}{6} \right] \left[\frac{1}{\epsilon} + \log \frac{k_\perp^2}{\bar{\mu}^2} \right] - L^2 + \frac{67}{18} - \frac{\pi^2}{3} \right\} \times$$

$$\int_{x,y,x',w'} e^{-ik \cdot (w'-x)} \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2} \left[U^{ab'}(x') - U^{ab'}(w') \right] N_c \left[U^{af}(y) - U^{af}(x) \right] \rho^{b'}(x') \rho^f(y)$$

(The overall factor is $\frac{g^4}{8\pi^4} \cdot \frac{1}{k^+} \cdot \frac{k^+}{2\pi} \cdot (-\pi) = -\frac{g^4}{16\pi^4}$, with the k^+ cancelling, so the UV term has no leftover k^+ .)

And the remainder, which needs no regulator:

$$\left. \frac{d^3 N_{2b}^{\text{NLO}}}{d^3 k} \right|_{\text{fin}} = \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+} \int_{\Lambda}^{k^+-\Lambda} \frac{dp^+}{2\pi} \int_{x,y,z,x'} e^{-ik \cdot (w' - \bar{\eta} z - \eta x)} \left\{ \left[(1.16) \text{ braces} \right] - C_{\text{UV}}(\eta) \mathcal{W} \frac{1}{(z-x)^2} \right\} \times (\text{colour})$$

at $d_{\perp} = 2$. Verified UV convergent — $u^2 \times \langle \cdot \rangle_{\hat{u}}$ as $u = z - x \rightarrow 0$:

$ z - x $	(1.16) braces	counterterm	difference
10^{-2}	1.34485×10^{-2}	1.34483×10^{-2}	1.6×10^{-7}
10^{-3}	1.34483×10^{-2}	1.34483×10^{-2}	1.6×10^{-9}
10^{-5}	1.34483×10^{-2}	1.34483×10^{-2}	1.7×10^{-13}

The counterterm carries the phase $e^{i\bar{\eta}k \cdot (z-x)}$, not 1, so it also matches the original's $\frac{1}{z^2}$ tail behaviour at large $|z|$ and the subtracted integral converges at both ends.

5. What is left, and one thing worth re-checking

Part 2a is yours, and the two-sided plus prescription is the right tool: it has poles at *both* $p^+ \rightarrow 0$ (from the $-\frac{\bar{\eta}}{\eta}$ block) and $p^+ \rightarrow k^+$ (from $-\frac{\eta}{\bar{\eta}}$), the companion $h(\eta) = \frac{e^{-i\eta\kappa}}{\eta A + \bar{\eta} B}$ is finite at both ends, and the result carries $2 \log \frac{k^+}{\Lambda}$. And 2a has no UV, so the ordering is free.

Worth re-checking in (1.10), the first part. Your second and fourth terms there are written in q^+ with the phase $e^{-ik \cdot (w' - x)} e^{-ik \cdot (x - z) q^+ / k^+}$, i.e. obtained from the first and third by the relabelling $x \leftrightarrow z$, $q^+ = k^+ - p^+$. I verified for the second part that the transverse-times-phase factor *is* invariant under exactly that relabelling — but that the **colour structures are not** symmetric under $x \leftrightarrow z$. In (1.10) the colour factor $f^{adc} U^{db}(x) U^{ce}(z) \{\rho^e(y), \rho^b(y')\} - \dots$ is written *identically* in all four terms. If those terms came from relabelling, the colour indices and the y, y' labels have to be relabelled with them. Please check that, or tell me and I will.

Checks: `src/rowB2_2b_uv.py` (the master integral against the Bessel quadrature, I_0 and I_1 in sympy and by quadrature, the subtracted remainder), `src/rowB2_check.py` (the six blocks, prefactor, p^+ weights, UV residue), `src/rowB2_split.py` (the 2a/2b split), `src/rowB2_identity.py` (the fusion identity), `src/rowB2_pplus.py` (the two-sided plus prescription for 2a).